Arnav Govindu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.41
Narayana Junior College	Class XII (TS-BIE)	2023	98.20%
The Hyderabad Public School, Begumpet	Class X (ICSE)	2021	93.50%

EXPERIENCE

AI Engineer - Kynos Logic

- Developed a native C++/MFC desktop application for processing million-point LiDAR LAS datasets, implementing AI-based classification, tiled feature extraction, OpenGL visualization, OpenMP-based processing, post-processing, statistical reporting with natural language filtering, and classified LAS export.

POSITIONS OF RESPONSIBILITY

Chassis engineer - Aeolus Racing, Formula SAE Team, IIT Hyderabad

- Represented IIT Hyderabad at Formula Student SAE Supra 2025, contributing both as a chassis engineer and as part of the business presentation team.
- Helped design and deliver the team's business presentation to judges, highlighting technical and financial aspects of the project.
- As a senior core member, mentor junior team members and guide chassis development through help with design decisions, technical reviews, and knowledge transfer within the domain.

SKILLS

- **Languages:** C++, Python, C, Java, SQL
- **Libraries/Frameworks:** PyTorch, Pandas, Numpy, Matplotlib, Scikit-learn
- **Tools:** Visual Studio, LaTeX, SolidWorks, ADAMS, ABAQUS, MATLAB, Excel, Git/Github

RELEVANT COURSES

- **Mathematics:** Calculus - I, Calculus-II, Transform Techniques, Statistics, Ordinary Differential Equations
- **Programming and AI:** Introduction to Programming, Introduction to Modern AI, Machine Learning and its Applications, Automation Lab
- **Mechanical Engineering:** Kinematics and Dynamics of Mechanisms, Heat and Mass Transfer, Manufacturing Science, Fluid Mechanics, Solid Mechanics, Thermodynamics, Engineering Mechanics
- **Computational & Design:** FEM & CFD, Modelling & Simulation, Fundamentals of Robotics, Machine Drawing & Solid Modelling, Computer Integrated Manufacturing
- **Electives:** Analysis & Design of Composite Structures, Marine Robotics, Industrial Engineering & Operations Research
- Foreign Languages: Spoken Japanese Basics

PROJECTS

Chassis Design & Fabrication - Aeolus Racing, Formula Student Team, IIT Hyderabad

- Designed the spaceframe chassis focusing on structural rigidity, driver ergonomics, and weight optimization.
- Performed FEA (Finite Element Analysis) for static and crash load cases to ensure safety and reliability.
- Iterated chassis geometry based on simulation results to meet competition standards and performance goals.
- Collaborated with team members to fabricate the chassis using welding techniques.

Emotion Recognition from Audio (Deep Learning Project)

- Developed a deep learning system to classify human emotions from speech audio clips by combining sound patterns and prosody features.
- Built a fusion model integrating a CNN (spectrogram features) and an MLP (prosody features: pitch, loudness, tempo), achieving 90.3% accuracy, a significant improvement over individual models (60% and 67%).
- Designed preprocessing, training, and evaluation pipelines in Python (PyTorch, Scikit-learn).

Simulation of a Crane-based mechanism

Course Project - Kinematics and Dynamics of Machinery

- Performed degree-of-freedom analysis, forward kinematics using homogeneous transformation matrices, and MATLAB based computation of end-effector displacement, velocity, and acceleration.
- Validated results through CAD modeling in SolidWorks and dynamic simulation in ADAMS, comparing trends with symbolic MATLAB outcomes.

Finite-Volume CFD Solver for Heat and Fluid Flow | Python

- Developed a Python-based 2D finite-volume solver for progressively complex heat and fluid-flow problems, implementing TDMA, upwind discretization, and pressure-velocity coupling using SIMPLE.
- Validated solver behaviour through convergence analysis and temperature, pressure, and velocity field predictions under multiple boundary conditions.

Physics-Informed Neural Networks for Computational Elastodynamics

- Implemented a composite mixed-variable PINN framework in **PyTorch** for transient elastodynamics, predicting displacement and stress fields using governing PDEs, constitutive relations, and boundary/initial conditions without labelled training data.
- Replicated the plate-with-hole benchmark from published research, comparing predicted stress distributions and transient von Mises response against FEM/reference results while investigating loss weighting and collocation sampling effects on convergence and accuracy, improving predicted peak von Mises stress from ~1.7 to ~2.6 kPa toward the ~3.3 kPa FEM reference.

CERTIFICATIONS

- **IBM Data Science Specialization**

EXTRACURRICULAR

- Represented IIT Hyderabad in the quiz team at Inter IIT Cultural Meet 2024 and 2025, hosted at IIT Patna and IIT Kanpur respectively.
- I enjoy playing badminton and table tennis, and also participate in competitive FPS video games like Counter-Strike.
- Helped organize Cryptex - The annual cryptography based online treasure hunt event at IIT Hyderabad.
- I practice speedcubing with 3x3 to 5x5 cubes.
- I play acoustic guitar, and I like playing songs from bands like Radiohead and Coldplay.